



IEEE Student Branch

AY 2025-26

# **COMPUTER AIDED ELECTROMAGNETIC FIELD ANALYSIS**

**(2025-2026)**

**Event Report**

By IEEE SVCE Student Branch

Conducted on 4<sup>th</sup> and 5<sup>th</sup> February, 2026

svce.ac.in



**BROCHURE**

**CAEFA WORKSHOP**

**IEEE SVCE Student Branch**  
Presents  
**A Two-day technical workshop on**  
**COMPUTER AIDED ELECTROMAGNETIC**  
**FIELD ANALYSIS**  
&  
Guest lecture by



**Dr. Deepak Arumugam**  
Group Lead, Innovation and Development  
Versuni India Home Solutions Ltd.

**Dr. KR Santha**  
IEEE Counselor, Vice Principal,  
Dean Research and Professor

- 📍 Software laboratory, EEE Department
- 📅 4<sup>th</sup> and 5<sup>th</sup> February, 2026
- 🕒 9am to 3pm
- ✓ Reg fee: ₹100 - SVCE students  
₹ 200 - Other college students

**MENTORS**

- 1) Dr. R.Karthikeyan / Professor / EEE
- 2) Mr. V.Karthikeyan / Assistant Professor / EEE
- 3) Dr. S.S.Sethuraman / Associate Professor / EEE
- 4) Mr. D.Gnanavelu / S/W Lab Instructor / EEE



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# BRANCH

## COMPUTER AIDED ELECTROMAGNETIC FIELD ANALYSIS (2025-2026) REPORT

The IEEE SVCE Student Branch successfully organized a two-day workshop on “Computer-Aided Electromagnetic Field Analysis” on 4th and 5th February, 2026 at the Software Laboratory Department of Electrical and Electronics Engineering. The workshop was exclusively conducted for the students of the Electrical and Electronics Engineering (EEE) and Electronics and Communication Engineering (ECE) Departments. The workshop aimed to enhance student’s understanding of electromagnetic field concepts and introduce them to modern computational tools used for field analysis. A total of 30 students from both departments actively participated in the program, showing keen interest in learning practical simulation techniques. The program provided students with valuable exposure to software-based analysis using Magnet helping them bridge the gap between theoretical knowledge and real-world engineering applications.



invocation song was delivered to mark the formal opening of the session



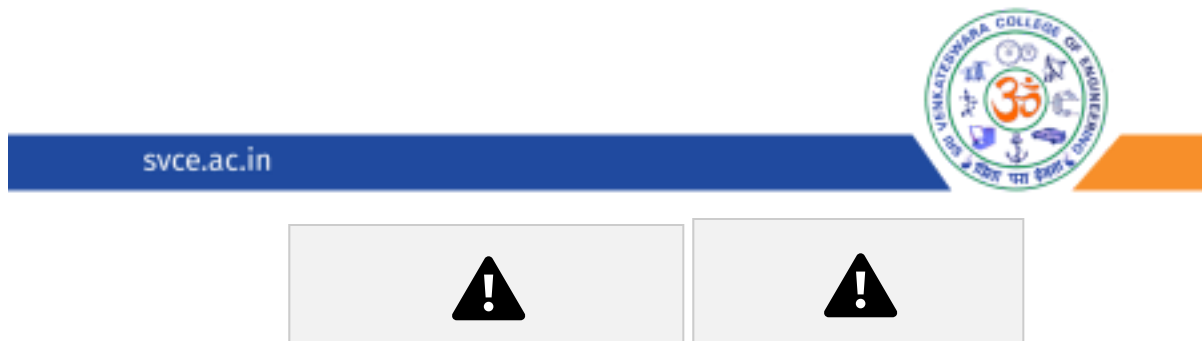
The welcome address was delivered by Mr. T R Hemachander, Vice Chair. He welcomed the Guest

Lecturer Dr. Deepak Arumugam, and also extended warm greetings to Dr. K. R. Santha, IEEE Counselor and Vice Principal, Dr. Sudhakar K. Bharatan, Head of the Department of Electrical and Electronics Engineering, Dr. S. G. Bharathidasan, Assistant Head of Department of Electrical and Electronics Engineering along with the mentors Dr. R. Karthikeyan, Professor, Dr. S. S. Sethuraman, Associate Professor and Mr. V. Karthikeyan, Assistant Professor from the Department of Electrical and Electronics Engineering. He acknowledged the presence of faculty members, organizing committee members, participants, and highlighted the importance of the workshop as a hands-on, application-oriented learning experience that integrates Finite Element Analysis, Magnet software, to help students understand the practical application of electromagnetic field concepts. He concluded by emphasizing IEEE's commitment to knowledge sharing, innovation, and technical development, encouraging students to strengthen their technical knowledge and explore advancements in electromagnetic field analysis.

Dr. K. R. Santha, Vice Principal and IEEE Counselor, addressed the gathering and reflected on the journey of IEEE within the institution. She briefly traced its evolution, noting that the IEEE activities were initially associated primarily with the EEE and ECE departments before coming under centralized coordination. Her remarks highlighted the steady growth and institutional support that strengthened the Student Branch over the years. She further emphasized that the theme of the workshop, Computer-Aided Electromagnetic Field Analysis, is closely aligned with core academic subjects of both Electrical and Electronics Engineering and Electronics and Communication Engineering. She underlined the importance of mastering such core concepts, as they form the foundation for advanced technical learning and professional competence.

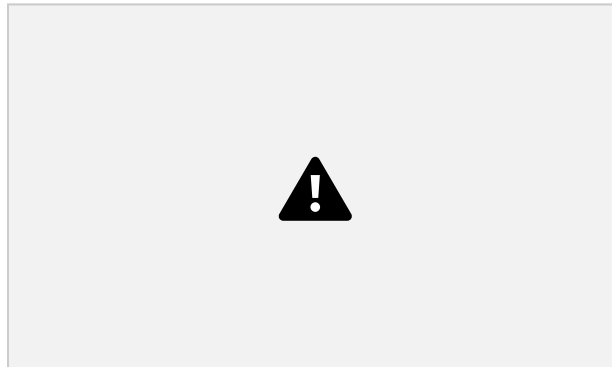


**Insightful Address by Dr. K. R. Santha**



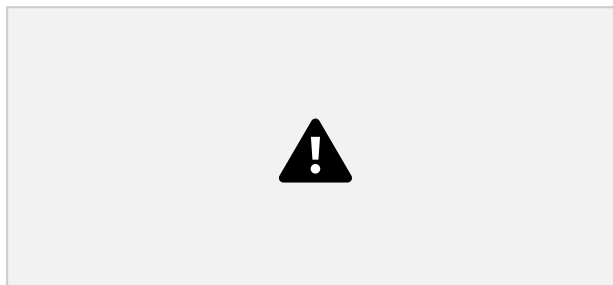
Following this, Dr. Sudhakar K. Bharatan, Head of the Department of Electrical and Electronics

Engineering, highlighted the importance of IEEE membership in supporting students' academic and professional development. He emphasized that IEEE offers access to valuable technical resources such as journals, conferences, and platforms like IEEE Spectrum, which help students stay updated with technological advancements, and encouraged them to actively engage with these resources and maintain their membership.

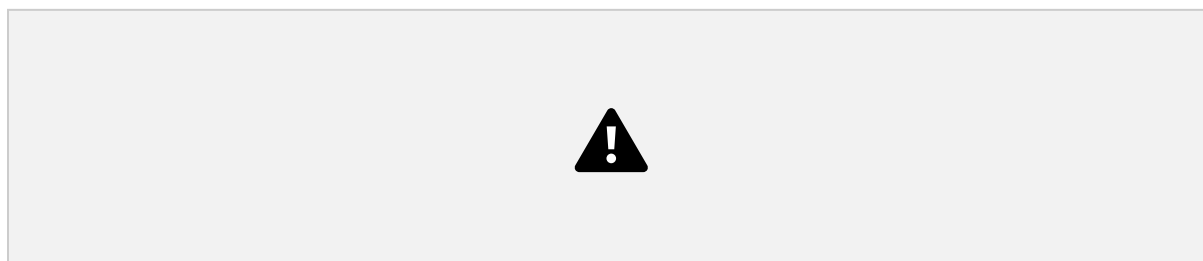


**Dr. Sudhakar K Bharatan, addressing the gathering**

The Chief Guest, Dr. Deepak Arumugam, was formally introduced by Mr. Nithish D. R., Event Head. Dr. Deepak Arumugam was introduced as Group Lead – Innovation and Development at Versuni India Home Solutions Ltd, along with his distinguished academic background and extensive industrial expertise in innovation-driven product development. His significant contributions to core engineering domains were highlighted, underscoring his impact in the field. This formal introduction seamlessly set the stage for his lecture, paving the way for an insightful session on advanced engineering tools and methodologies.



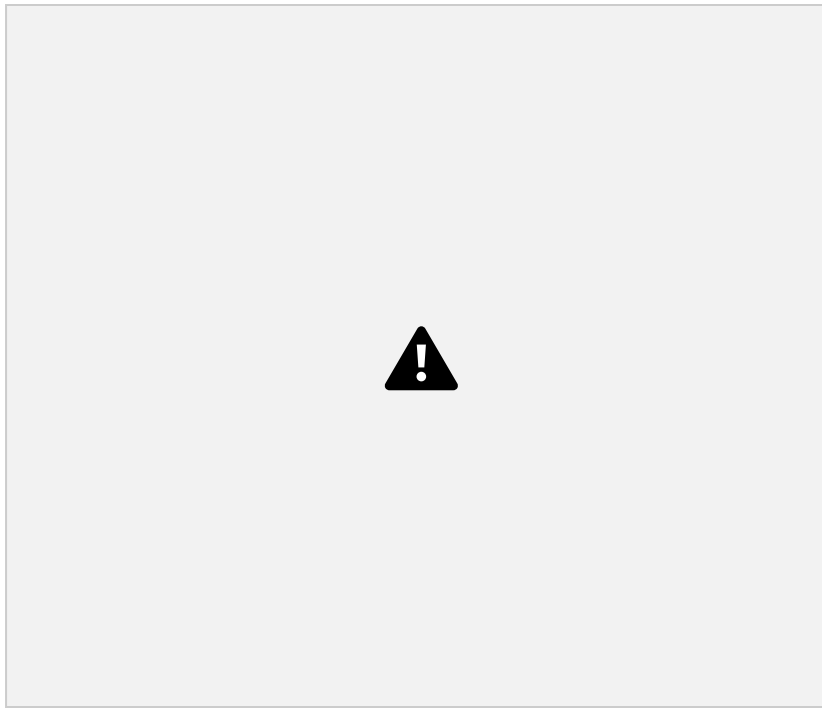
**Chief Guest Introduction by the Event Head**



Dr. Deepak commenced his lecture by outlining the career opportunities available in the field of Computer-Aided Design (CAD). He discussed the growing demand for proficient design engineers across automotive, electrical, and consumer electronics industries, while presenting a balanced perspective on the benefits and limitations of CAD in modern engineering workflows.

Expanding on current technological trends, Dr. Deepak addressed the growing influence of Artificial Intelligence in engineering. While acknowledging its transformative impact, he emphasized that core engineering domain particularly those involving electromagnetic field analysis and electrical machine design still require strong human expertise and cannot be fully replaced by AI. He stressed the importance of building a solid foundation in core subjects, noting that such knowledge is crucial not only for technical interviews but also for long-term professional excellence in engineering careers.





**participants with PPT**

**Guest Speaker engaging**



In his address, Dr. Arumugam presented a comprehensive overview of Computer-Aided Engineering (CAE) tools, with particular emphasis on Finite Element Analysis (FEA). He detailed the application of FEA-based methodologies in Electrical and Electronics Engineering (EEE) and Electronics and Communication Engineering (ECE), focusing on the analysis and optimization of electrical machines.

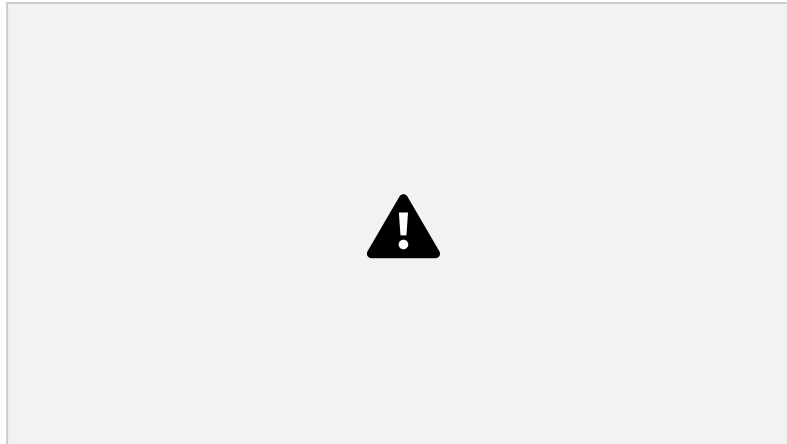
Critical aspects such as electromagnetic field distribution, thermal characteristics, and performance validation were discussed, demonstrating the importance of simulation-driven design in modern engineering practice.

Dr. Deepak further shared insights from a recent industrial project undertaken by his team involving the design and development of hub motors. This discussion provided students with practical exposure to contemporary engineering challenges and industry-oriented problem-solving approaches. Addressing emerging technological trends, he emphasized that core engineering roles involving electromagnetic field analysis and electrical machine design cannot be entirely replaced by Artificial

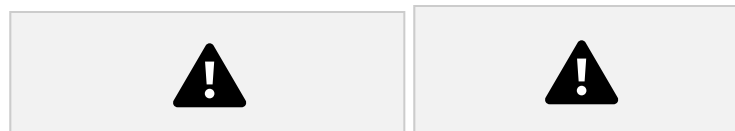
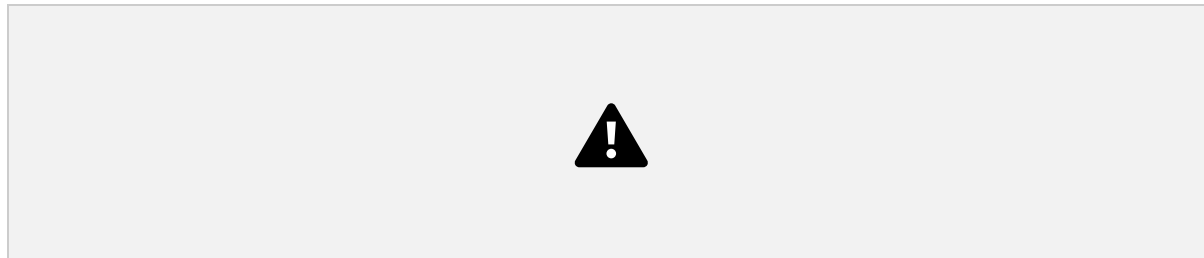


Intelligence.

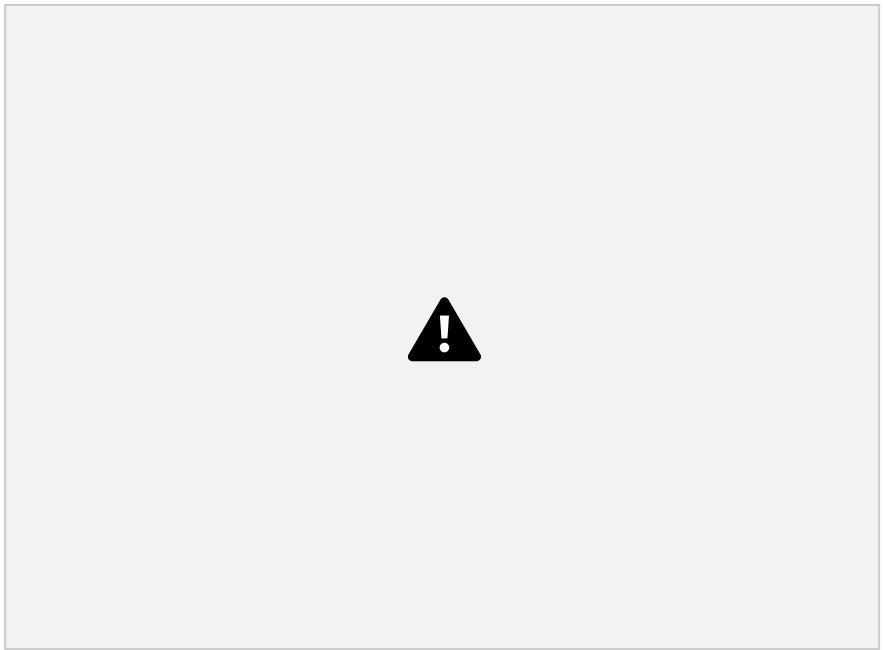
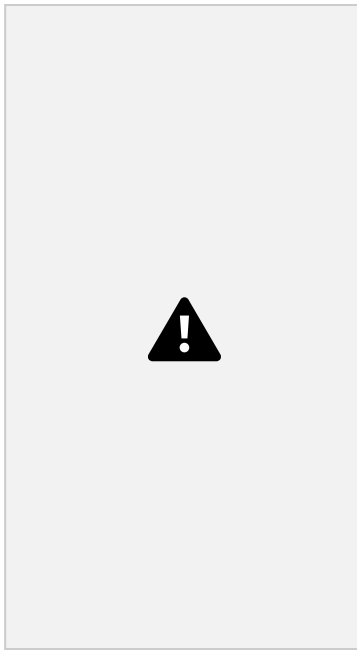
The session concluded with guidance on potential project domains such as motor modeling and PCB design, followed by an interactive discussion where students inquired about Battery Management Systems (BMS) and hub motors. The lecture effectively combined technical depth with professional insight, leaving a lasting academic and motivational impact on the participants.



**Dr. Deepak addressed the growing influence of Artificial Intelligence**

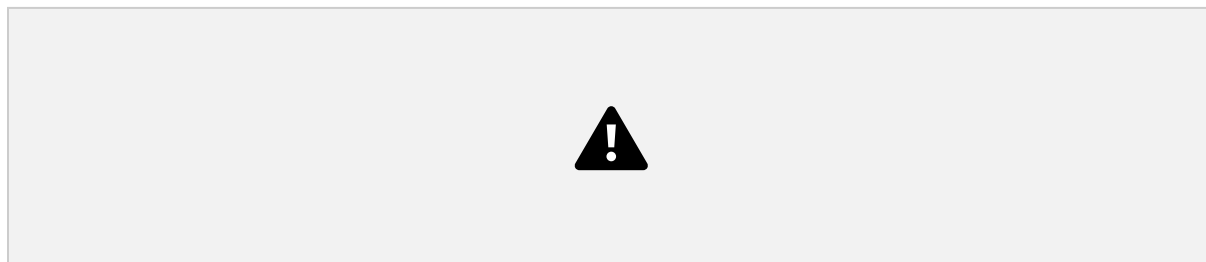


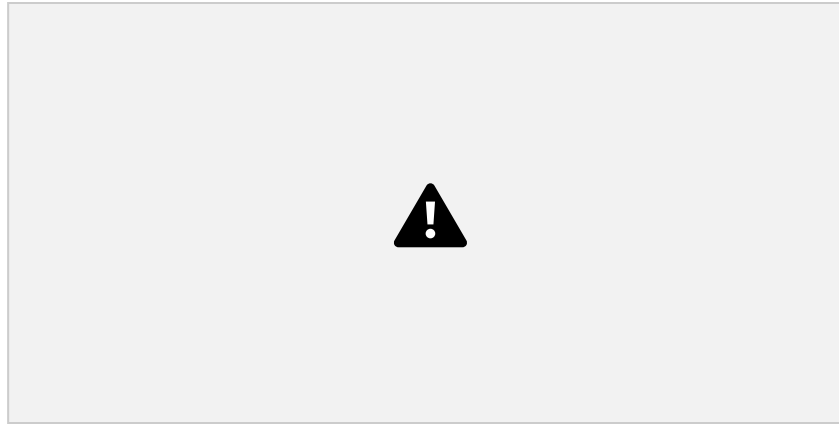
Following the address by the Guest Lecturer, a technical session was conducted by the Mentors, Dr. R. Karthikeyan and Dr. S. S. Sethuraman, who delivered a presentation on Computer-Aided Design (CAD). The session introduced the fundamental concepts and significance of CAD in modern engineering practices. During the presentation, the mentors explained several theoretical aspects associated with design and analysis, including Neumann's Condition, highlighting its relevance in engineering computations and boundary value problems commonly encountered in simulation-based design processes.



**Technical session by Dr. R. Karthikeyan**

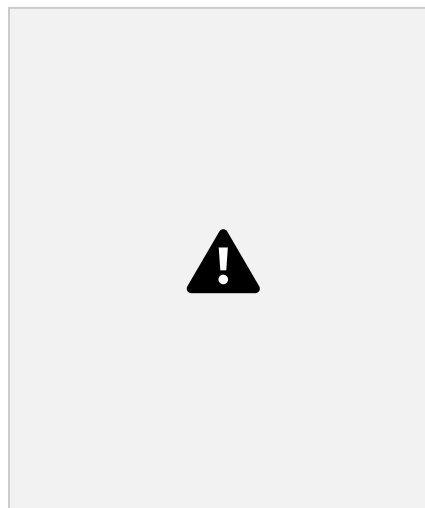
The session further elaborated on how such theoretical principles are incorporated within CAD supported analytical tools to improve the reliability and accuracy of engineering models. Through the presentation, the mentors provided an overview of how CAD facilitates efficient design development, visualization, and modification of engineering components.





**Technical demonstration by Dr. S. S. Sethuraman**

Following this, the discussion extended to the conceptual stages involved in CAD-based design processes. The mentors explained the structured workflow typically followed in design development, beginning with conceptual design, progressing to digital modeling, and subsequently leading to detailed design refinement using CAD software. The importance of iterative design evaluation and digital representation in achieving precise and optimized engineering solutions was also highlighted.



**Technical segment by Dr. R. Karthikeyan**

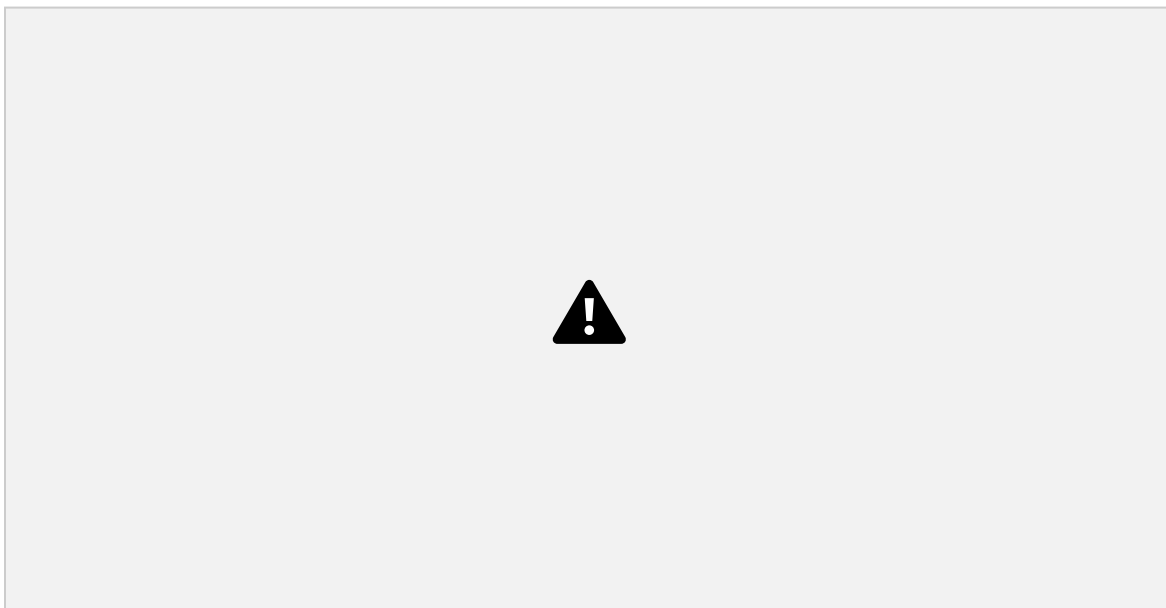


In addition to these topics, the session briefly addressed the broader applications of CAD across various engineering domains, including drafting, product design, simulation, and manufacturing

integration. The mentors emphasized how CAD technologies assist engineers in creating accurate digital models while enabling efficient analysis and design improvements.

Following the technical session by the mentors, a hands-on training session was conducted for the participants. During this segment, participants were given the opportunity to apply the concepts discussed earlier through guided practice using CAD software. The mentors demonstrated essential modeling procedures and provided assistance as participants familiarized themselves with the software interface and basic design commands.

The practice session continued with active participation, enabling the attendees to gain preliminary exposure to the practical aspects of computer-aided design. With the completion of these activities, the first day of the workshop was brought to a close.

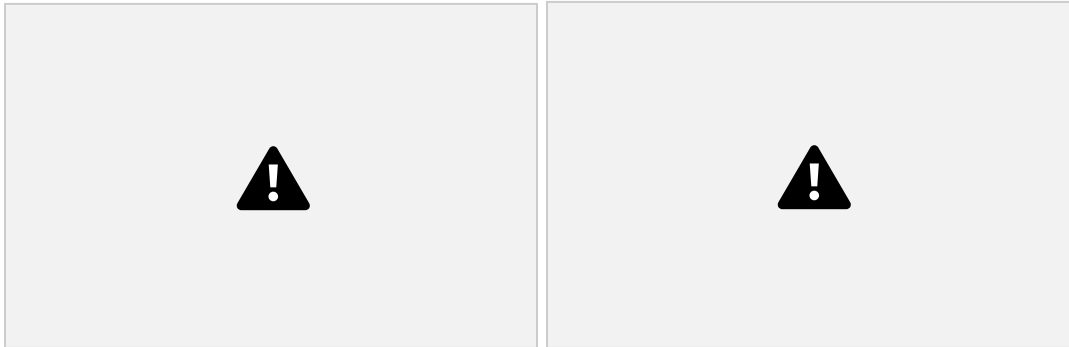


**Participants actively engaged in Practical Implementation**



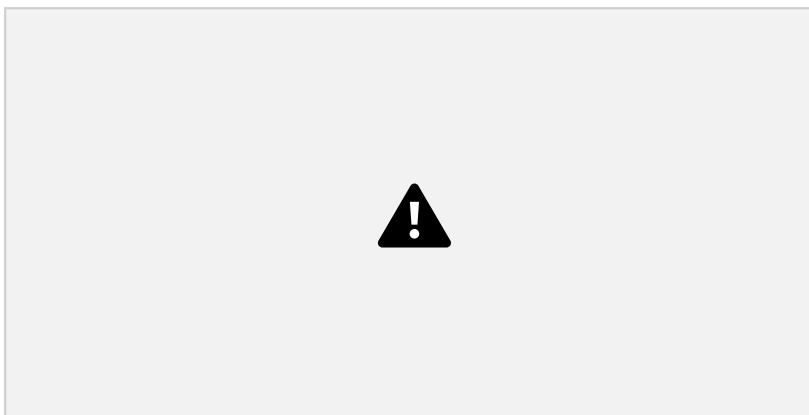
On the second day of the workshop, the session commenced with demonstration of selected tutorials aimed at providing participants with a deeper understanding of simulation and analysis using

computational tools. The mentors presented a tutorial on the Single-Phase Transformer, explaining its modeling aspects and demonstrating how computational tools can be used to study transformer behaviour and performance characteristics.



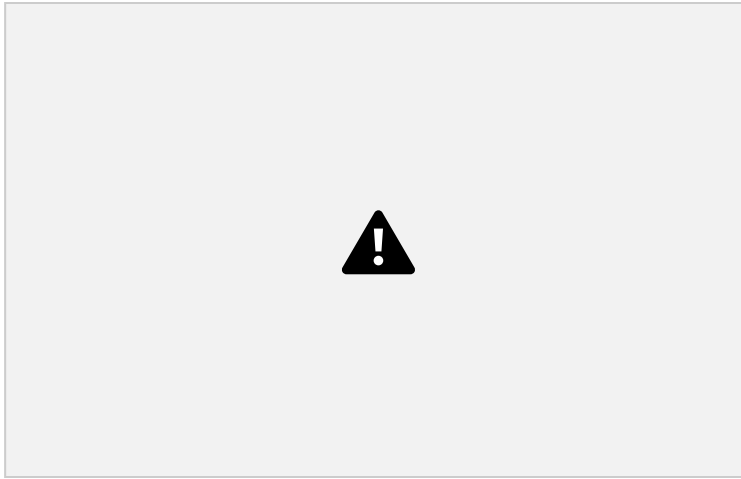
#### **Dr. R. Karthikeyan Presenting the Technical Content to Participants**

This was followed by a session on Electrostatic Field Analysis using the MATLAB PDE Tool, where the mentors demonstrated the process of analysing electrostatic field distributions through the MATLAB environment. The session highlighted the use of the PDE toolbox for defining boundary conditions, modeling field regions, and visualizing simulation results, thereby illustrating the practical application of computational methods in electromagnetic analysis.



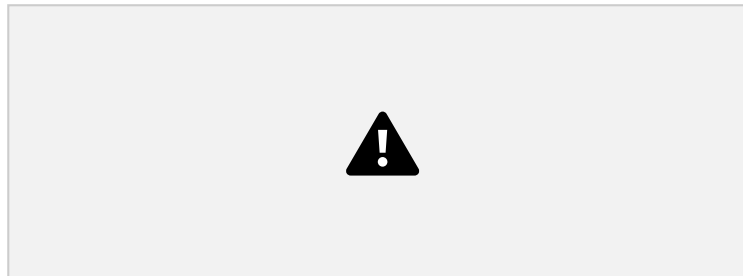
#### **Mentor's active guidance to participants**





**Dr. S. S. Sethuraman guiding students during the Hands-on Session**

Subsequently, a hands-on practice session was conducted, during which participants were encouraged to implement the tutorials that had been demonstrated earlier. Under the guidance of the mentors, the participants worked with the software tools to perform the modeling and analysis procedures discussed in the sessions. The mentors provided step-by-step guidance and clarifications, enabling the participants to understand the practical workflow involved in computational analysis and design. During this session, the participants actively engaged in executing the simulation procedures, exploring the software interface, and applying the relevant commands required for modeling and analysis. This practical engagement allowed them to strengthen their conceptual understanding while simultaneously developing familiarity with the operational aspects of the tools used. The session therefore served as an important component of the workshop, reinforcing the theoretical discussions through structured practical implementation.



**Participants actively engaged in hands-on training**

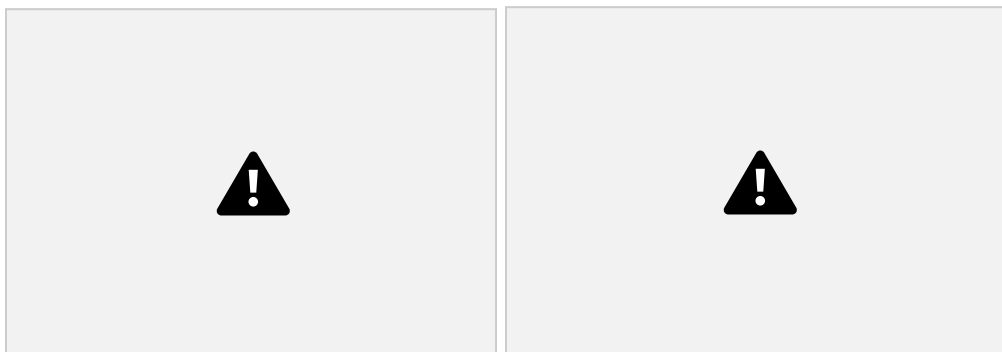


The valedictory session of the Computer-Aided Electromagnetic Field Analysis Workshop commenced in the presence of distinguished dignitaries, Dr. Sudhakar K. Bharatan, Head of the Department, Electrical and Electronics Engineering, and Dr. S. G. Bharathidasan, Assistant Head of the Department, Electrical and Electronics Engineering, along with our faculty mentors Dr. R. Karthikeyan and Dr. S. S. Sethuraman, and faculty co-ordinators.



#### **Valedictory Ceremony with the HOD, AHOD and Mentors**

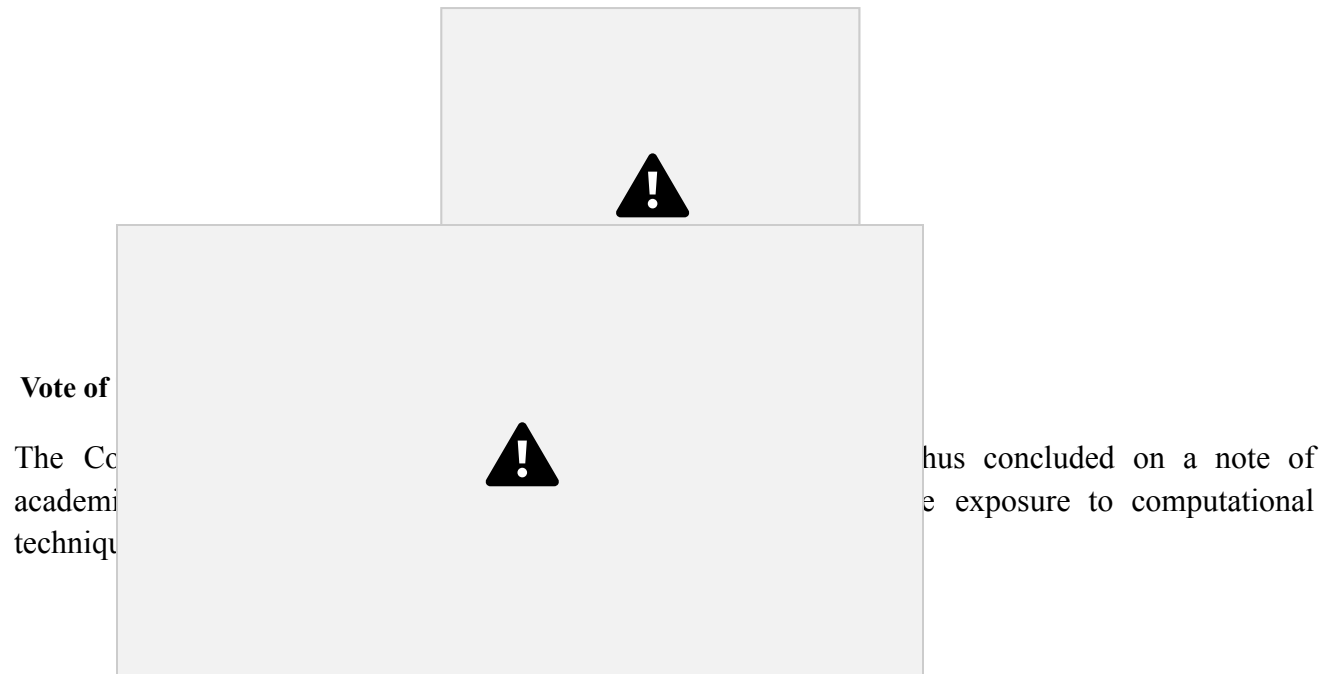
The certificates were presented to all the participants by Dr. Sudhakar K. Bharatan, Dr. S. G. Bharathidasan, and Dr. R. Karthikeyan, in recognition of the student's active involvement throughout the workshop. A feedback session was subsequently conducted, during which participants provided their valuable feedback and constructive insights regarding the workshop.



#### **Formal ceremony where students receive certificates and share valuable feedback**



The session concluded with the Vote of Thanks delivered by Mr. E.Shriram, Social Media Executive, who expressed sincere gratitude to Dr. K. R. Shantha, IEEE Counselor and Vice Principal, for her continued support and encouragement. He thanked Dr. Sudhakar K. Bharatan, Head of the Department, faculty mentors, and faculty co-ordinators for their guidance and support, and extended his heartfelt thanks to all the participants whose enthusiasm and active engagement contributed significantly to the success of the workshop.



**A group photo capturing a memorable moment during the IEEE event**

